

Bin the Butt: A Design for Behavioral Change towards Cigarette Butt Litter

4208 words

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ABSTRACT

This design case investigates the behavior of littering cigarette butts on the floor by student smokers on the smoke-free TU/e campus. Since the implementation of The New Tobacco Act in August 2020, all ashtrays and smoking bins have been removed from the TU/e campus leading to an increased number of cigarette butt litter on floor. This increase in cigarette butt litter is a huge concern for the environment including birds and animals. This design case focuses on designing interactions and interventions to induce a behavior change using theoretical models and frameworks like the Theory of Planned Behavior (TPB) and the Trans-Theoretical Model (TTM), as theory learned in the course DDM110 Design for Behavior Change

Authors Keywords

Behavioral change; cigarette butt litter; smoking; litter reduction; attitude

INTRODUCTION

For the course Design for behavior change, we decided to choose a very peculiar design challenge, targeting the behavior of throwing cigarette butts on the floor after people are finished smoking, instead of throwing the cigarette butts in a bin or an ashtray. It may

seem as a simple straight forward behavior, but what makes our challenge unique is that we are mainly focusing on students who smoke on TU/e campus, and as of 1st of august 2020, the TU/e is a no – smoking campus due to “The General Measure of Governance for smoke-free school sites”, drawn up by the Ministry of Public Health (VWS)[6]. This means that all school sites (from primary school to higher professional education/ university) must be completely smoke free.

The implementation of this new law led to the removal of ashtrays around campus, but this did not stop the students (18 years or older) to continue smoking on campus. The students have ended up making unofficial smoking spots on campus and the lack of ashtrays has led to an increased amount of cigarette butt litter all around campus, especially in the “unofficial smoking areas”.

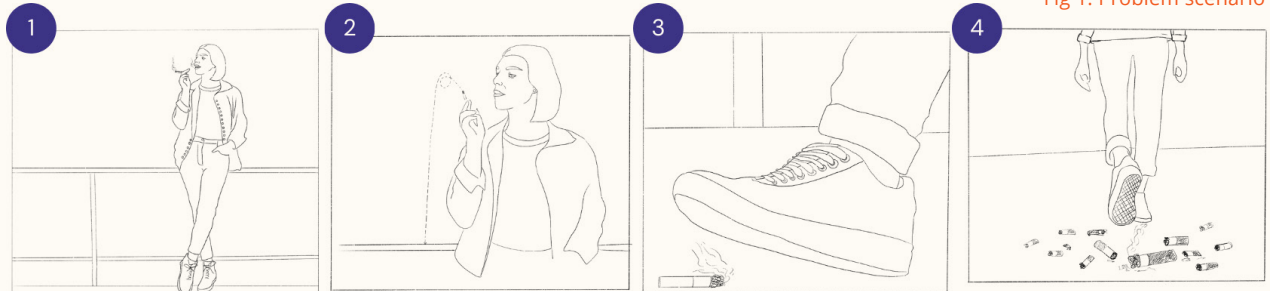
Littering of cigarette butts on the ground showcases a behavioral belief that cigarette butts are not trash . This poses a threat to the environment. Cigarette butts have filters which

are soaked in the tar produced by the burning of tobacco, filled with many toxins that are harmful for the environment, and they are also not bio degradable [1, 7]. Netherlands has a very windy climate, which often blows these cigarette butts into the soil and very often birds and small animals like bunny rabbits (who live and are often seen around the Tu/e campus) often try to eat the littered cigarette butts or get choked on them as well.

As designers we decided to take up this challenge to determine what is causing this behavior and select the appropriate theoretical model to design an intervention to change this behavior. Our goal is to achieve a cigarette butt litter free campus or at least a decrease in the number of cigarette butts on the floor.

To determine what is the reason for this specific behavior, we decided to further define the problem, by putting the specific target behavior in context and analyzing the key determinants of behavior change using tools like the COM-B analyses.

Fig 1: Problem scenario



DESIGN PROCESS & CONCEPT DISCRIPTION

COM-B analysis

From the specified challenge, we start by identifying what would need to change to achieve the new behavior. This process begins with a COM-B analysis, this is a framework introduced by Michie et al. [8] that suggests that three elements are necessary to perform a certain behavior (B): capability (C), opportunity (O), and motivation (M). By conducting a COM-B analysis, we can understand the behavior and learn what aspects need intervention to change this behavior. The behavior we are targeting in this analysis is throwing cigarette butts into bins instead of on the ground. Figure 2 shows the COM-B analysis, which will be briefly discussed below.

There is no need for change in physical capability, as one can see and walk towards a bin to dispose of cigarette butts properly. However, there is a significant need to change psychological capability, as people lack the awareness of the consequences of their behavior.

There are already several opportunities to dispose of cigarette butts, but there could be improvements in accessibility and visibility. Mainly the social opportunity calls for change, as there is not enough social accountability currently for calling out others on improper disposal of cigarette butts on the floor.

In both ways, there is a need for motivational change, as smokers often have a reflex to throw their cigarette butts on the ground quickly without considering the consequences. This also shows that smokers do not go through the deliberate thought process of picking up or properly disposing of butts.

The COM-B diagnosis shows that interaction designers can contribute to the psychological capability, social opportunity, and the reflective

	What needs to happen for the behavior to occur?	Is there a need for change?
Physical Capability	See, walk, move towards a dustbin or ashtray	No because people have the capabilities
Psychological Capability	Awareness towards consequences - Aesthetic and environmental factor	Yes, lack of awareness of additives, aesthetic and environmental factors, they don't see butts as trash
Physical Opportunity	Small disposal items or locating nearest bin	Yes, the campus is a non-smoking area and the places where they smoke there is no smokers infrastructure
Social Opportunity	Takes one person to make you feel ashamed/call you out of your behaviour and someone that does the behavior	Yes, people need to call out others for their bad behaviour
Reflective Motivation	Having the thought process of picking up or throwing away the cigarette butts	Yes, there needs to be a motivation to throw those cigarettes into a bin or ashtray
Automatic Motivation	Reflex of not having something dirty on your hands, so you throw it	Yes, have a other method to keep it until you find somewhere to throw it
COM-B diagnosis	Interaction designers can contribute to the Psychological Capability, Social Opportunity, Reflective Motivation and Automatic Motivation.	
Determinants	Attitude towards Cigarette-Butt Litter: beliefs that cigarette butts are litter, and norms such as the perception of whether it is acceptable to dispose of cigarette butts improperly. Social stigma of smoking in a non-smoking place. Beliefs about the biodegradability of cigarette butts. Motivation of changing habit. Strength of habit.	

Fig 2: COM-B analysis

and automatic motivation. The figure shows which determinants we found, in this design case we are targeting attitude towards cigarette butt litter and the perceived control [4] of being able to dispose of one's cigarette butt.

Behavior Change Strategy

From the COM-B analysis, we can link the componens to the intervention functions as seen in Figure 3. As there was no need for change in physical capability, this row remains unmarked, but the other rows are relevant for our target behavior. Through this matrix [8], we decided that education and enablement would be the most suitable, because education would change attitude and psychological capability, while enablement would change physical opportunity. Table 1 shows the overview of intervention functions and behavior change techniques for our design case.

Enablement goes beyond education or training

Fig 3: Intervention functions matrix

		Intervention Functions								
		Education	Persuasion	Incentivization	Coercion	Training	Restriction	Environment restructuring	Modelling	Enablement
COM-B Components	Physical capability									
	Psychological capability									
	Physical opportunity									
	Social opportunity									
	Automatic motivation									
	Reflective motivation									

Intervention Function	Behavior Change Techniques	Relevant Theory
Education	Increasing knowledge or understanding, e.g., by providing information that explains the importance and consequences of the behaviour	Theory of Planned Behavior: Education can strengthen attitude and beliefs
Enablement	Increasing means/reducing barriers to increase capability or opportunity	Theory of Planned Behavior: Making it easier to perform a specific behavior can strengthen perceived behavioral control

Table 1: Behavior Change Strategy

by providing tools, resources, and motivational support that targets barriers to behavioral change, fostering self-efficacy and motivation. Contrary to environmental restructuring, enablement can also include social strategies and incentives that empower smokers to change their behavior towards cigarette butt litter.

The Theory of Planned Behavior aligns with our design case, as one of the COM-B determinants we target is attitudes, education could affect this. On the other hand, enablement would influence the perceived behavioral control. Therefore the behavioral intent towards is affected as well, ultimately leading to the desired behavior.

Concept Ideation

The ideation process started with three different concepts (Fig 4): an integrated ashtray, an art wall, and voting bins.

The integrated ashtray would encourage smokers to put their cigarette butt in the small ashtray that is integrated into their pack of cigarettes. As smokers always take their pack with them, this would be an accessible solution, however, this would require cigarette pack manufacturers to change the design, which is not feasible within the design case. Giving smokers a portable ashtray instead would not be effective, as we

have seen from personal experience with the Peukensee ashtrays [9], smokers forget to take this item with them.

The art wall would encourage smokers to place their cigarette butts in an artwork, as it may spark interest in smokers and motivation to participate. This would also influence smokers' attitudes towards cigarette butts through visual and creative appeal. However, such an intervention would only move the problem to another location but not actually clean the area, and attract insects. This would undermine our goal to focus on environmental sustainability.

The voting bins would encourage users to properly dispose of their cigarette butts by voting on a social dilemma. This would offer immediate feedback as they can see that others participated as well and have voted, which could enhance participation. Next to this, as the social dilemmas would be humorous, the joys of social interaction could motivate smokers to put their butts into the bin, instead of on the ground.

When reflecting back on the definition of the behavior change strategy, the most promising concept was the voting bins, however, this concept still needed an iteration to foster both education and enablement.



Fig 4: First concept sketches

Bin the Butt



Fig 5: Bin the Butt concept

This led to the Bin the Butt campaign, which consists of three elements:

1. An Instagram campaign which aims to educate smokers on the consequences of their behavior, alternative interventions, and motivation to action (Appendix A). This platform could easily reach the target user in a low-threshold, humorous manner.
2. Routes for smokers to take during their smoke breaks. These would enable them to walk past bins to dispose of their butts, while simultaneously leading them off-campus. The spillover effect of these routes are the benefits for one's health to go on a walk, which adds to the motivation to action.
3. Voting bins and signs along the routes. The voting bin intervention enables smokers to properly dispose of their butts in an interactive manner. The signs along the routes increase awareness about their wrong behavior and its consequences, and nudges them to change their behavior.

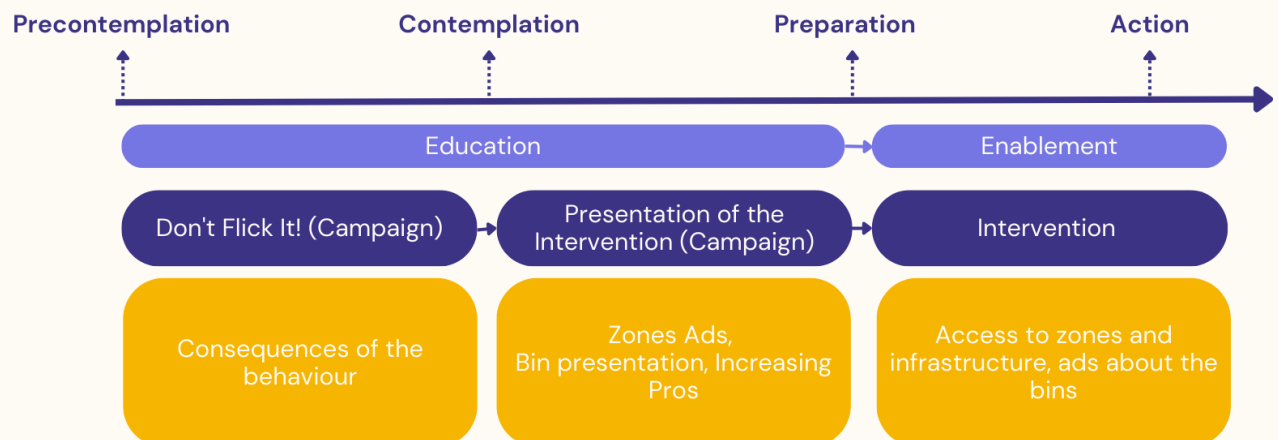
After analyzing our insights from the pilot evaluation, which will be discussed in a later section, we iterated our intervention once more. This was mostly based on the insight that the

participants did not have enough time in their smoke break to take a walk outside of campus. Therefore, **Bin the Butt** changed from a three-component intervention to a two-component intervention (Figure 5):

1. The **Instagram campaign** remains the same.
2. **Legal smoking areas with the smoking infrastructure** close to campus replace the routes, as the participants indicated that they do walk towards a safe location nearby campus where smoking is allowed. Instead of placing the bins and signs along the routes, we only place them in designated areas, this would increase their participation in behavior change, as the placement aligns with their current smoking location, instead of having to lead them somewhere.

We use the Trans-Theoretical Model to analyze the steps that are needed to change the attitude of the target user, according to (Figure 6). The Bin the Butt educational campaign would allow the user to go from precontemplation, to contemplation. Enabling them to walk the routes would prepare them for action, while the bins would actively encourage them to take action within their behavior.

Fig 6: Intervention mapped to Trans-Theoretical Model stages of change



THEORETICAL RATIONALIZATION FOR DESIGN CONCEPT

The intervention is based on the Theory of Planned Behavior (TPB). We choose this theory due to its capability to help explain why individuals behave or not behave in a certain way. TPB takes into account their attitudes, subjective norm and perceived control. It provides a framework for designing interventions that aim to change behavior by targeting these specific elements [3].

The educational campaign targets the behavioral belief of people that cigarette butts are not considered litter. Research has shown that this attitude is a key reason that smokers often fail to dispose of cigarette butts correctly [5]. The campaign aims to shift these beliefs (Figure 7), changing attitudes towards disposal and increase the intent to take part in the desired behavior of disposing butts in a correct manner.

The second mechanism of change Figure 8, addresses environmental constraints affecting behavior. As the campus is a non-smoking area, smokers have the tendency and need to hide their actions [1,3], which is further amplified by their perceived lack of control in proper disposal. A non-smoking area like the campus provides a lack of accessible bins and ashtrays. The

intervention focuses on promoting the desired behavior by building additional infrastructure just outside campus to increase access to proper disposal. This will increase perceived control as it will become easier for smokers to dispose of their cigarettes without getting caught.

For this intervention, we decided to target behavioral beliefs and control beliefs. The education campaign raises awareness about the environmental impact of litter and conveys the pros and cons of doing said behavior, focusing on behavioral belief. On the other hand, positioning dedicated smoking areas with the required infrastructure, near the campus, focuses on control beliefs. The accessibility of will help enable smokers to act responsibly.

There are three implementation considerations based on TPB:

Although, interventions targeting behavioral, normative, or control beliefs, can be successful in changing attitudes, subjective norms, and perceived control. Consequently, increasing behavioral change. That is not always enough for an individual to change their behavior. If individuals cannot act on their intentions, the intervention will be ineffective. It is then important to strengthen the link between

intention and behavior, which can be achieved by creating specific implementation intentions [3].

Furthermore, to adapt the study to the target demographic, open-ended elicitation can be used to identify the key factors that influence behavior in that specific context. In addition, to allow us to better understand how attitudes, perceived control and intentions drives the behavior change it is recommended to do a prospective study design, where we measure the factors at one point and then track the actual behavior later [3].

IMPLEMENTATION DESIGN RATIONALE

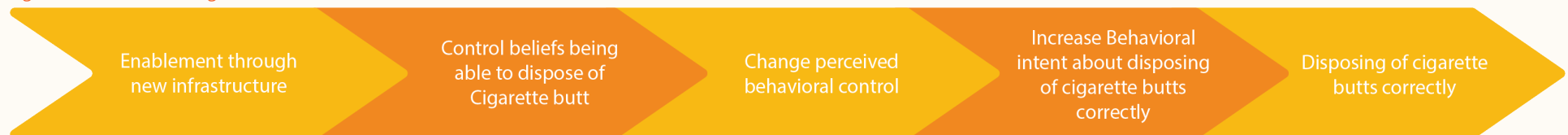
In this project, we used Education and Enablement techniques and combined persuasion based on Fogg's Functional Triad framework to shape our design concepts and processes, and used these techniques to "persuade" users to handle cigarette butts responsibly. Here, we used Fogg's three important roles: Medium, Tool and Social actor in our design. We will discuss the application of these three roles in our design in detail below (Figure 9).

For the Education technology, we used Instagram as a "Medium" to raise awareness quickly and effectively. Through campaigns, posts, and

Fig 7: Mechanism of Change on TPB for Education



Fig 8: Mechanism of Change on TPB for Enablement



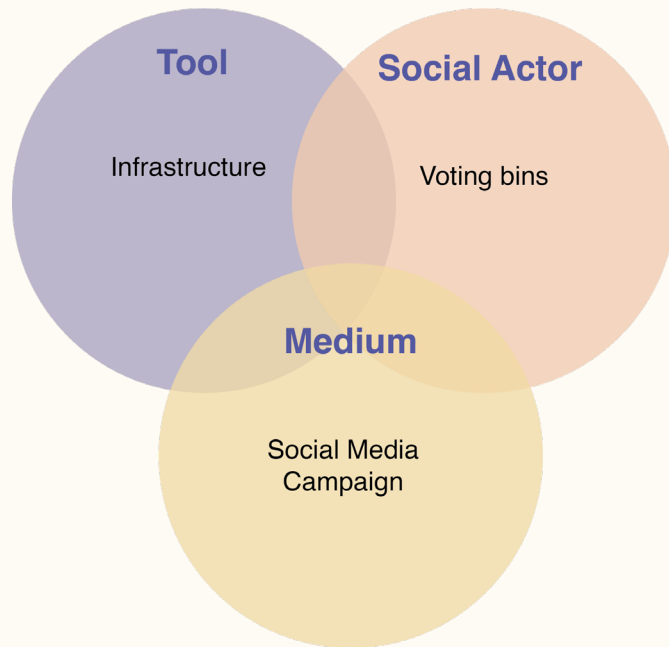


Fig 9: Functional Triad Bin the Butt

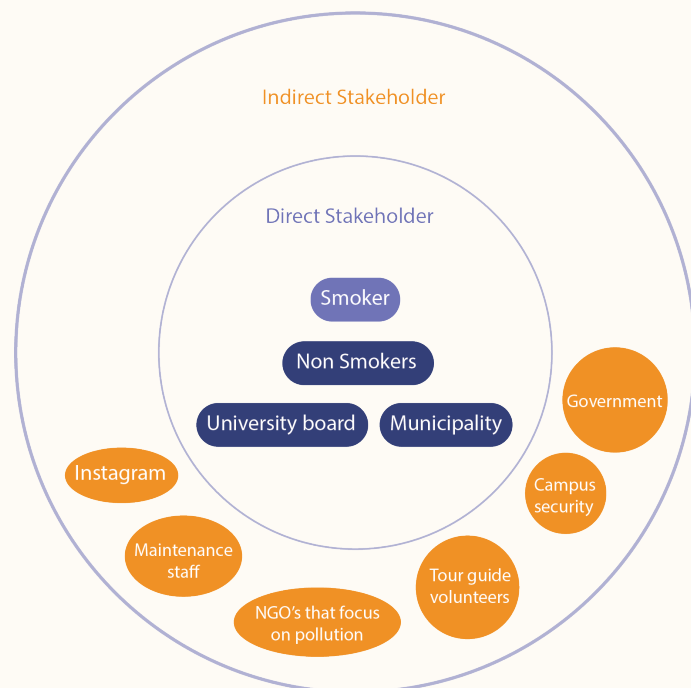


Fig 10: Stakeholder Mapping Bin the Butt

stories, we highlighted the environmental harm and risks to animals caused by cigarette butt littering, encouraging users to take responsibility for proper disposal. Acting as an educational medium, these resources deliver clear and impactful messages that gradually build users' understanding of the environmental consequences of littering. This awareness encourages users to think about the outcomes of their actions, motivating them to adopt more responsible behavior.

In the Enable technology, our design focuses on removing barriers that may prevent users from properly disposing of cigarette butts. Here we chose the smoking area and prompts sign as "Tools" to persuade, and these elements provide a direct and convenient way for users to dispose of cigarette butts. In this case, the user will not give up this correct behavior because it is impossible to find and the route is complicated.

The role of "Social Actor" is reflected in the voting bins, through which users are invited to participate in collective activities with their cigarette butts. In this regard, the voting bins provides an immediate interactive experience, allowing users to "vote" by placing cigarette butts in the bins, and users can also see their own contributions and the choices of others. When placing cigarette butts in these bins, users feel that they are part of a common effort, which fosters a sense of community and responsibility, and allows them to experience a sense of pride in doing a small thing, but it can affect the environment.

These design elements are expected to support behavior change effectively by addressing users' cognitive and social motivations while reducing obstacles to desired behaviors. The Education raises cognitive awareness, enabling users to appreciate the long-term impacts of littering

on the environment, thus laying a foundation for voluntary behavior change. Enablement makes responsible behavior more accessible by reducing physical and psychological barriers. Through the interactive voting bins, which act as Social Actors, we transform the disposal process into a participatory and interesting social decision-making activities. Compared to approaches relying on signs or fines alone, this method promotes voluntary participation and strengthens positive behavior norms, increasing the likelihood of lasting change.

When considering alternative approaches, such as simple signage or punishments, we found that these options lacked the interactive feedback required to reinforce social norms effectively. Punishments, while there are significant short-term results, this does not address the problem at its root [2]. Negative reinforcement can sometimes cause resistance rather than a genuine commitment to change. By combining Education, Enablement, and Fogg's Functional Triad, our approach provides a more balanced solution. Positive guidance and interactive incentives encourage voluntary engagement, helping users develop long-term behavior through ongoing interaction and reinforcement. This design better aligns with users' psychological needs, avoids the downsides of punitive measures, and supports sustainable behavior change through repeated social interaction.

ETHICS ANALYSIS OF DESIGN CONCEPT

Design for Behavioral Change is a field that aims to get a target group to adopt desired behaviors and alter existing ones with use of persuasive techniques from behavior theories. The target group is not always aware of the persuasive influence, which creates ethical question that must be addressed with careful analysis.

Through stakeholder mapping, direct and indirect stakeholders were identified (Figure 10). By examining the pains and gains of these groups, we defined four key moral values:

1. Accessibility

By making infrastructure and information accessible allows smokers to engage with the intervention and access smoking areas without stigma. It would encourage them to dispose their cigarette butts properly, and foster inclusivity and respect for individual choices.

2. Environmental Sustainability

Promoting responsible disposal through educational campaigns and cigarette voting bins, reduces litter and protects campus environments which upholds a clean and sustainable environment. This, in turn, would give a sense of environmental responsibility among smokers.

3. Respect

It is important to treat all parties involved respectfully. The intervention should not be stigmatizing to smokers or any other parties. This could be accomplished by providing non-judgmental spaces for them as well as respecting non-smokers' right to a smoke-free environment. This would reflect fairness and empathy.

4. Welfare

Well-being of the community should be held in high regard by minimizing the exposure to second-hand smoke. This value supports public health by placing smoking areas in low-traffic areas, which helps people avoid second-hand smoke. Designated smoking areas would lead to a generally smoke free campus that is clean from cigarette litter.

These values, however, reveals a potential conflict that could create ethical tensions. First, although the intervention does not focus on the habit of smoking, it might have an impact on it. By increasing accessibility to smoking areas, it might have the adverse effect of reinforcing smoking behavior which conflicts with the value of public health. This could potentially be an unwanted outcome, but predictable and unethical putting the fault to the design team responsible (Figure 11). To avoid this outcome, the design team should therefore reflect on the design of the smoking areas by using the help of stakeholders. The areas should provide the infrastructure needed but not encourage users to linger and become social hubs, by for instance having no sitting area.

Secondly, if these voting bins overflow or are poorly maintained, litter might actually increase. While the aim is to provide easy access, the intervention risks actually undermining environmental sustainability goals if the design is not effective. Again, these outcomes are predictable and unethical (Figure 12). To avoid this, the design team should collaborate with the municipality and University board to ensure proper maintenance of the infrastructure. In addition, the voting bins should be robust and easy to replace.

Lastly, placing smoking areas near non-smokers' walking paths leads to unintentional exposure, impacting health. The placement may or may not contribute to passive smoking which conflicts with the welfare value. Again, a reasonable and unethical outcome (Figure 13). To avoid it, the design team should organize a discussion with all stakeholders to democratically decide where to place the smoking areas by using data gathered about the popular and less popular locations around the campus.

Fig 11: Ethical Decision Tree 'More smoking'

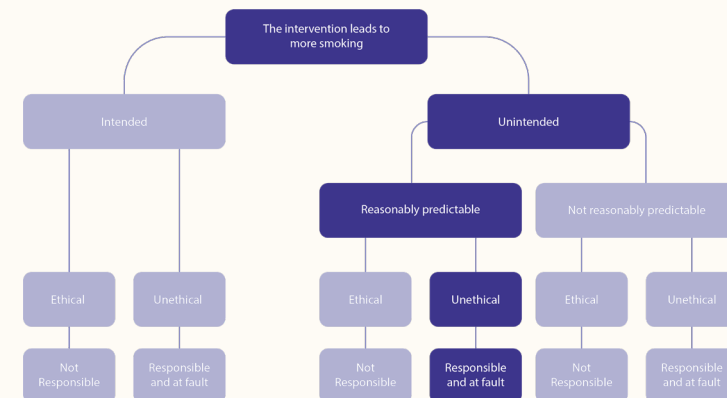


Fig 12: Ethical Decision Tree 'Nature litter'

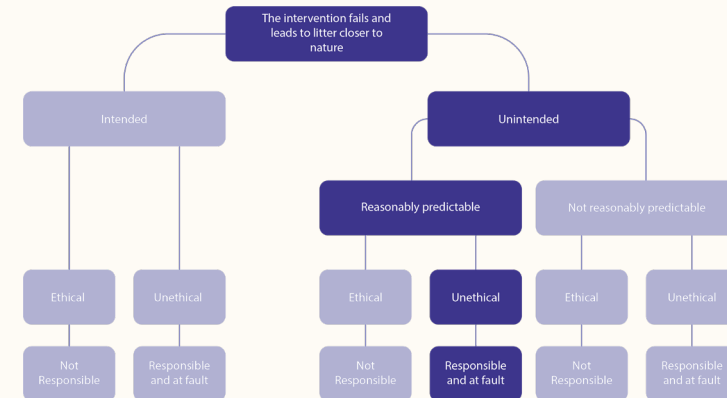
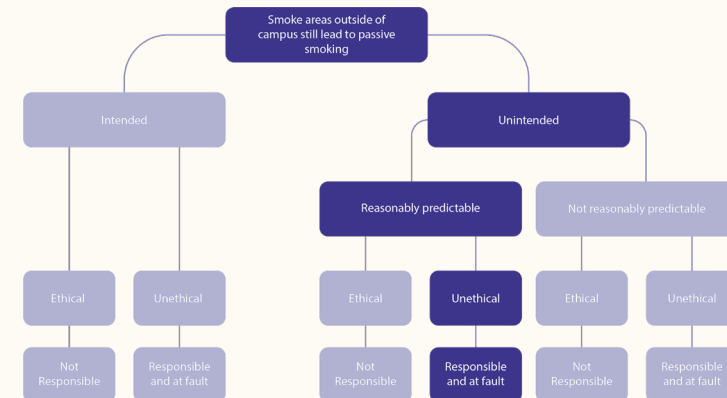


Fig 13: Ethical Decision Tree 'Passive smoking'



Ethical Guidelines

Our approach emphasizes empathy, transparency, and collaboration. Ethical tensions are addressed by engaging with stakeholders through interviews, focus groups and participatory design sessions (smoker and non-smokers). This will help us refine our intervention to make sure it aligns with our core ethical guidelines.

1. Promote Responsible Disposal (based on Environmental Sustainability value):

Prioritize sustainability by ensuring easy and responsible disposal of cigarette butts. This reduces environmental harm and creates a sense of accountability.

2. Ensure Respectful Interaction (based on Respect and Welfare values):

Develop non-stigmatizing solutions that consider the welfare of all parties, promoting respect for smokers while also respecting non-smokers' rights.

3. Prioritize Accessibility and Usability (based on Accessibility value):

Make interventions practical and easy to understand to encourage widespread adoption, ensuring the approach remains transparent.

By following these guidelines and decision-making processes, we aim to create a balanced, ethically responsible intervention that respects all stakeholders and promotes a healthier, more sustainable environment.

EVALUATION PLAN

To evaluate the designed interactions and interventions, specific evaluation plan needs to be laid out (Figure 14). This evaluation will consist of two participant groups:

- Intervention group

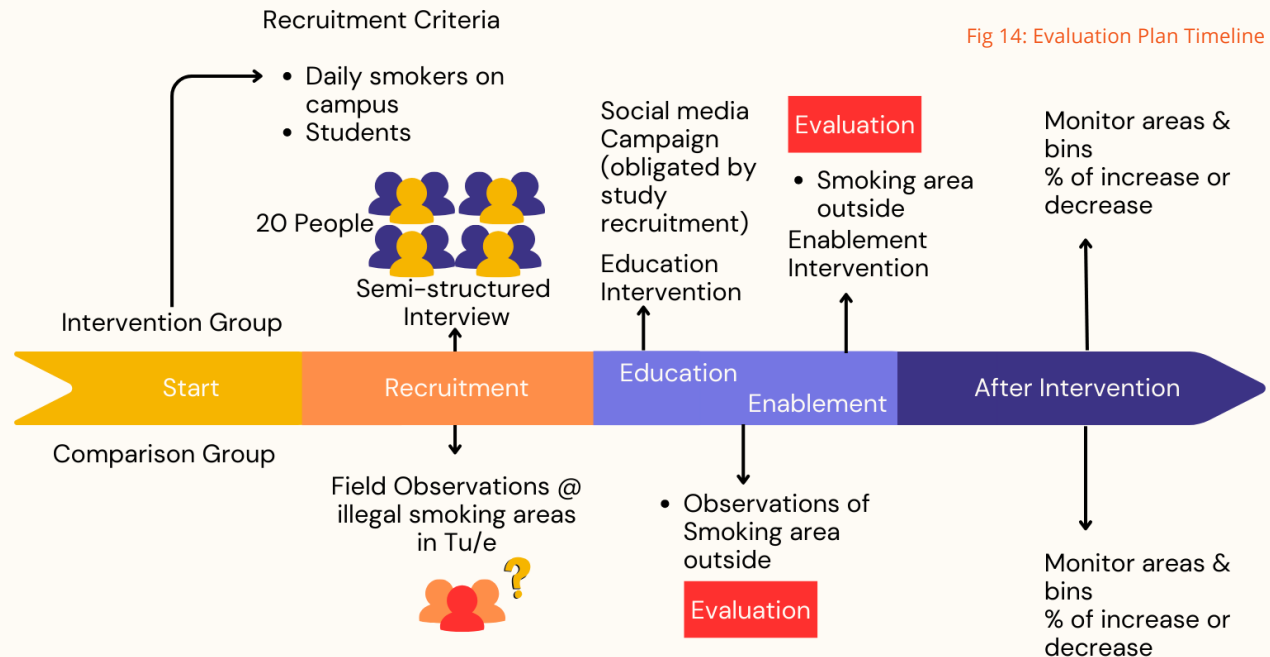


Fig 14: Evaluation Plan Timeline

- Comparison group

1. Intervention group

The intervention group will consist of participants recruited through semi-structured interviews, making sure that they are daily smokers on campus and are students at the TU/e campus. The main aim of the semi structured interviews is to gather information about the following:

- **Attitude:** questions will be asked to understand what their attitude is about littering of cigarette butts on the floor (prospective study).
- **Behavioral Belief:** Their current beliefs, do they consider cigarette butts as trash?
- **Stage of behavior:** The participants will be asked questions to determine which stage of change they are currently in (pre-contemplation, contemplation & preparation). Are they aware of the consequences of their current behavior? Do they plan to make any changes to their

behavior in the upcoming 6 months or not?

After the recruitment process, each participant will be asked to follow the Bin the Butt social media page on Instagram, on which they will view the educational posts about the consequences of littering cigarette butts on the floor.

During the evaluation period (6-12 months), each participant will maintain a smoke and trash journal. In this journal they will note down how many cigarettes they start the day with as soon as they reach TU/e campus, and after smoking where they disposed the cigarette butt.

At the same time, the physical intervention of the voting bins will be installed in the designated smoking areas, and at the end of each day the number of cigarettes butts in the voting bins will be counted.

This process will take place for 6-12 months. During this process, the research/design team will be continuously monitoring the statistics of the social media page, focusing on aspects like

how long a participant viewed a certain post, number of likes and reshares. Based on these statistics the social media posts may be slightly modified.

2. Comparison Group

The comparison group on the other hand will be the student smokers in the unofficial" smoking areas on campus. No interactions or intervention will be introduced to them. This group will simply be observed from a distance for the same duration of 6-12 months. At the beginning and the end of the day the number of cigarette butts on the floor will be counted.

Intervention Evaluation

At the end of the evaluation period, another round of semi structured interviews will be done with the intervention group. Questioning each participant about the same aspects as done during the recruitment process.

- Their new attitudes about littering cigarette butts
- Their new beliefs about cigarette butts as trash
- The stage of their behavior (pre-contemplation, contemplation and preparation)

Their new responses will then be compared to their initial responses.

The conclusion will be drawn with a summation of the data gathered through the smoke and trash journal and the total number of cigarette butts in the voting bins.

The results from the intervention group will then be compared with the observations and data (cigarette butts on the floor of unofficial smoking areas) gathered from the comparison group.

PROTOTYPE AND EMPIRICAL EVALUATION

Pilot Evaluation

As we aim to evaluate the enablement phases of Bin the Butt over a longer period to see the actual effects, it is not feasible to prototype the implementation of the zones with bins and signs within the timeframe of the course. For this reason, we chose to evaluate the education phase of Bin the Butt, which consisted of the social media campaign posts and stories for three consecutive days. The education phase also targets attitudes, which is one of our main determinants in this design case. We prototyped the bin and sign to be able to showcase it to the participants as well.

Three participants, who were all smokers, were asked to follow the Instagram account on the first day, see the posts and stories, after which they filled in a questionnaire (Appendix C). This questionnaire asked them to evaluate the campaign, routes, bins, and how this affected their attitude and motivation towards changing their behavior.

Interaction Evaluation

Based on the pilot evaluation (Appendix D), we did an interaction evaluation. From this we could conclude that the users did use the campaign as intended, as well as in the intended context: they engaged and viewed the posts through Instagram. However, the number of participants was limited to derive any real conclusions. The participants were willing to use the campaign intervention, but they indicated that they were more likely to and interested in engaging with positive and fun posts. The negative posts made them feel judged and refrain from behavior change.

DISCUSSION

In reflecting on this project, we found several areas that could be explored further. First, our

research data are based on literature reviews from American universities, however this may not be applicable to TU/e, as there are still cultural differences between Europe and the United States, so we may need to conduct elicitation interview to investigate their attitudes towards littering in Dutch universities. Additionally, much of our current approach is shaped by one team member's perspective as a smoker. Involving more TU/e student smokers would make our findings more well-rounded, especially for evaluating the social media campaign and interaction prototypes.

The short week-long intervention also limited our ability to see the full process of behavior change, which typically spans 6 months or more. A longer evaluation would give a clearer view of how students move from initial consideration to action. Observing behavior in both no-smoking and permitted areas, like the city center, would also be valuable, as littering still occurs in areas where smoking is allowed. We'd also need to clarify legal aspects, such as who is responsible for maintaining the voting bins and covering costs, especially if the installation is near campus but not directly on it. Finally, one participant noted that humorous signs left a lasting impression, so exploring different information styles, funny, grim, or educational, could offer insights into what resonates most with our target user.

INDIVIDUAL CONTRIBUTIONS

The Design case started with a discussion and brainstorming about different themes and interests. We then democratically decided and defined the chosen design case. All members of the team contributed evenly to this activity. Moreover, we collectively worked on the workshop's tasks and divided concept generation and description evenly. For the theoretical rationalization we worked together to define which theory would be best to base

our mechanism of change. However, Helena refined the mechanism of change and specified the theoretical rationalization.

Furthermore, the implementation design rational was done through a collective brainstorm where we together made the decisions, and Chenwei refined this part. Sanchayita and Chenwei identified the stakeholders through stakeholder mapping, that was further developed by Helena with the ethical analysis.

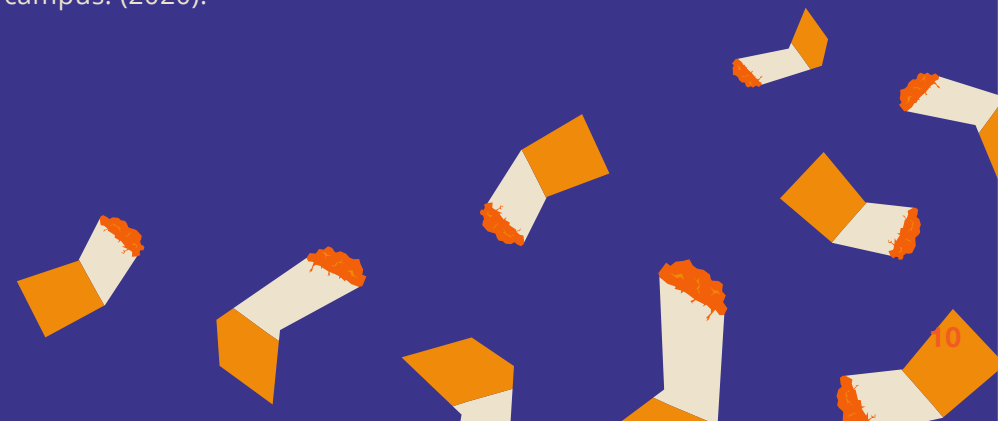
For the Ethical analysis, the group had first a discussion about ethics and our intervention. Helena worked out this part with the support of Sanchayita.

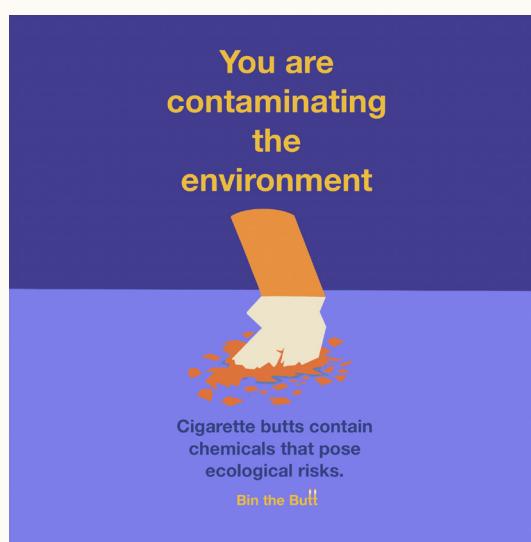
The Evaluation plan was mainly developed by Sanchayita and Helena, further refined as a group. Tess and Chenwei made the prototypes and Helena helped make part of the posts for the campaign as Tess did the rest of the campaign. Helena and Sanchayita recruited the participants for the test and Sanchayita made the Empirical evaluation questionnaire. Chenwei, Tess and Helena analyzed the data gathered from the participants. Finally, the analysis of the data was interpreted by Sanchayita and Helena, and further refined by Chenwei. The pictorial was designed and composed by Tess.

All in all, we believe that it was a balanced team, and many actions were taken collectively to ensure that Everyone achieved the learning goals of this course.

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- [7] 5 ways cigarette litter impacts the environment. Retrieved October 29, 2024 from <https://truthinitiative.org/research-resources/harmful-effects-tobacco/5-ways-cigarette-litter-impacts-environment>
- [8] The behaviour change wheel: A new method for characterising and designing behaviour change interventions | Implementation Science | Full Text. Retrieved October 29, 2024 from <https://implementationscience.biomedcentral.com/articles/10.1186/1748-5908-6-42>
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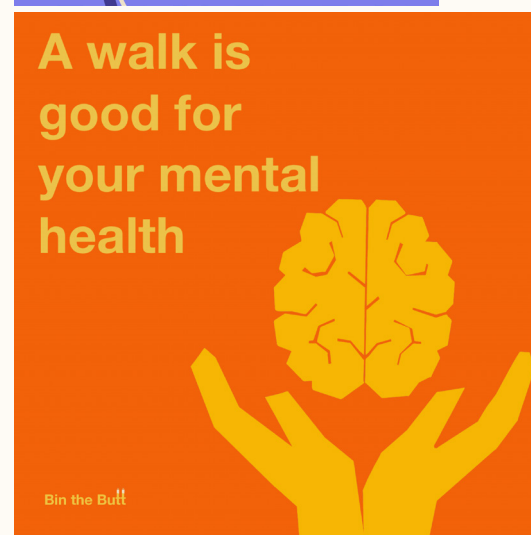
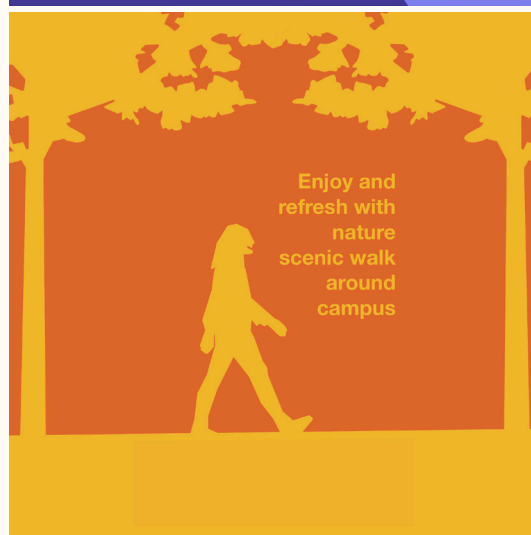




DAY ONE
Raise awareness around consequences



DAY TWO
Introduce the intervention elements



DAY THREE
State the pros (spillover effects) of intervention



First: get your 'design brief' clear.

E.g. losing weight, getting fit, improving school grades, etc. are not behavioural targets.

Behavioural targets: Eat 2000 calories daily, walk 10k steps per day, study 120' per day are behavioural targets.

Put the challenge in its context: The first idea for modifying a specific behaviour may need to be thought through:

- What is the real concern that calls for a behaviour change?
- Who is the individual, group or population involved in the behaviour?
- What behaviours influence the specific situation?
- Where do these behaviours take place? In which context?
- Who (else) is involved?
- What factors seem to influence these behaviours?
- Are there any interdependencies of behaviours? Specifically, look whether 'spillover' effects can be achieved.
E.g., going for a walk in the evening can mean sitting less in front of the TV and can reduce snacking as well!

1. Generate list of target behaviours to change

2. Assess each target behaviour in terms of

- **Impact** of behaviour change: how big an influence does a positive change of behaviour have upon the situation of concern?
- **Likelihood** of changing behaviour: think of capability, motivation and opportunity for people to change the behaviour.
- **Spillover**: how likely is it that changing this behaviour will bring positive/negative effects on other related behaviours
- **Measurement**: how easy is it to measure this behaviour?

Example scale by Michie: unacceptable/worth considering/promising/very promising)

3. Select the one that seems more promising (has best scores)

Target Behaviours	Impact	Likelihood	Spillover	Measurement	Total Assessmen
Not making the effort to look for a bin	Littering on the floor	High	Littering other small trash	Very Easy	Promising/ Unpromising/ Unacceptable
Smoking in a arear which is recluded or where you will not be judged	Smoking in an area not equipped to smoke	Low	Increase in passive smoking not being considerate to others	Very Easy	Promising/ Unpromising/ Unacceptable
Behaviour 3					Promising/ Unpromising/ Unacceptable
Etc.					Promising/ Unpromising/ Unacceptable

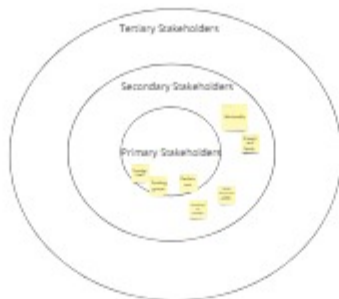
1.3 Specify target behaviour and its context

- **Who** needs to perform the behaviour? (characteristics)
- **What** do they need to do differently?
- **When** do they need to do this?
- **Where** do they need to do this?
- **How** often ?
- **Whom** with?

Target behaviour	Littering cigarettes
Who needs to perform the behaviour? (Characteristics)	Individuals with the habit of smoking and throwing their cigarettes on the ground.
What do they need to do differently?	They should look for a bin where they can through it away or keep it until they find one.
When do they need to do it?	When they have finished smoking or when they dicide to smoke (find somewhere where they can smoke and get read of their cigarettes
Where do they need to do it?	The smoking location
How often do they need to do it?	After every smoke
With whom do they need to do it?	Themselves and fellow smokers

Com-B components	What Needs to happen for the behaviour to occur	Is there need for change?
Physical Capability	See, Walk, move towards a dustbin or ashtray	No because people have the capabilities
Psychological capability	Awareness towards consequences - Aesthetic factor and Enviro factor	Yes, lack of awareness of additives, enviro factors, aesthetic factors
Physical opportunity	Small disposal items or locating nearest bin	No, these things are already out there
Social opportunity	Takes one person to make you feel ashamed of your behaviour/ to call you out and someone that does the behavior	Yes, people need to call out others for their bad behaviour
Reflective Motivation	having the process of picking up or throwing the cigarettes	Yes, there needs to be a motivation to throw those cigarettes
Automatic Motivation	Reflex of not having something dirty on your hands	Yes, have a other method to keep it until you find somewhere to throw it
Com-B diagnosis	write here a summary of what needs to change in terms of the Com-B elements	
Determinants	Write here what are the measurable quantities that we need to change - we shall later align them to the chosen theories	

COM-B Model Components		Definitions
Capability	Psychological Capability	Knowledge and psychological skills, strength, or stamina to engage the necessary mental processes.
	Physical Capability	Physical skills, strength, or stamina.
Motivation	Automatic Motivation	Automatic processes involving emotional reactions, desires (wants and needs), impulses, inhibitions, drive states and reflex responses.
	Reflective Motivation	Reflective process involving plans (self-conscious intentions) and evaluations (beliefs about what is good and bad)
Opportunity	Social Opportunity	The opportunity afforded by interpersonal influences, social cues and cultural norms that influence the way that we think about things, e.g., the word and concepts that
	Physical Opportunity	The opportunity afforded by the environment involving time, resources, locations, cues, physical ‘affordance.’



Write
your own
values

Assignment (see how to be graded on a scale from 0 to 100)

Perform a **value chainholder analysis** for your design case. List stakeholders, define who benefits and who loses, examine goals and issues in terms of **values**. Select and discuss at least four **core values** relevant to your design case. One of these can be called "substory." Motivate your selection. Describe the meaning of each core value and its relevance for the case and different stakeholders. Include a "dialog" & "terroir".

Assignment (to be included as a section in the report)

In one or more **paragraphs**, describe possible **unresolved** ethical issues/conflict in everyday life of your concept.

Reflect on your attitude and approach with respect to ethical concerns in your design work.

Formulate and include at least three relevant and nontrivial [ethical principles or moral principles](#) for your team, reflecting your intention and responsibility as designers. Assess the consequences of your design decisions for yourself, or for others, based on those principles.

2. Stakeholder analysis

[Frederman 2013] **Value Sensitive Design**

1. Define each underlying condition or values, or context of issue
2. Identify Critical and Potential Stakeholders
3. Identify the values and interests for each Stakeholder Group
4. Map/Identify and determine any **competing values**
5. Develop a Conceptual Framework for all Stakeholders
6. Identify the **Key Value Conflict**
7. Propose a Work Coordination Plan (How Do we Organizational Services)

Arak
almost
there

Stakeholder analysis

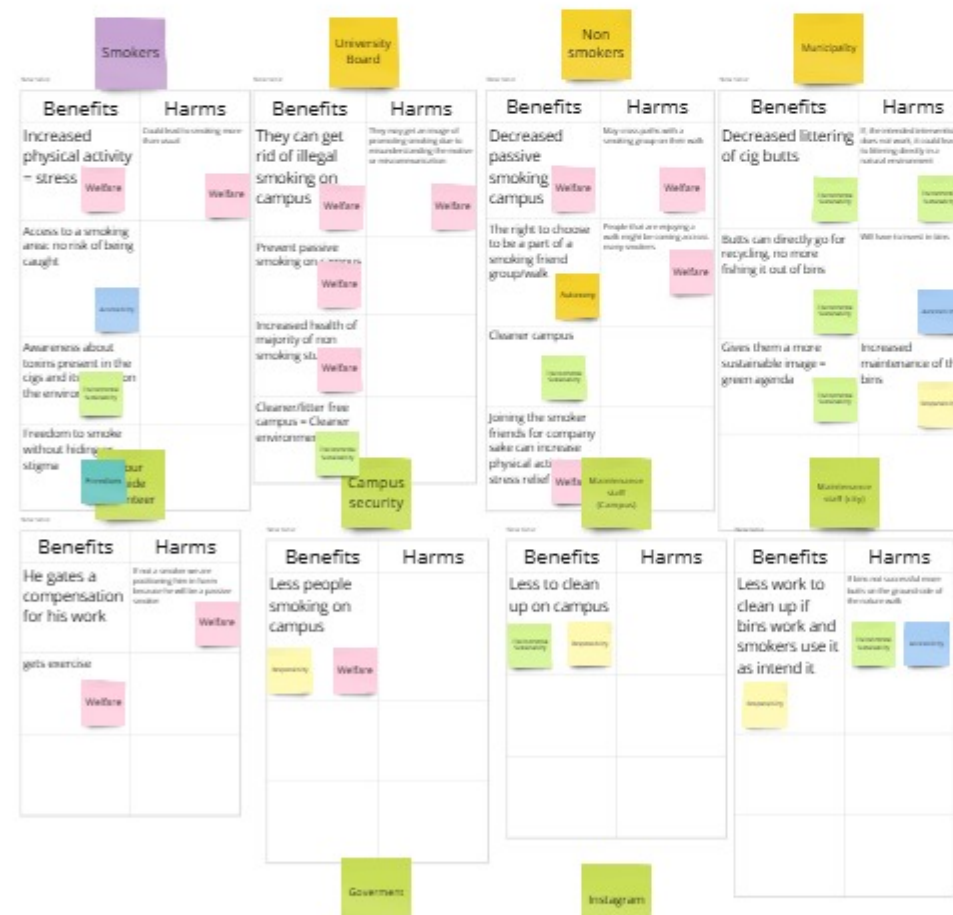
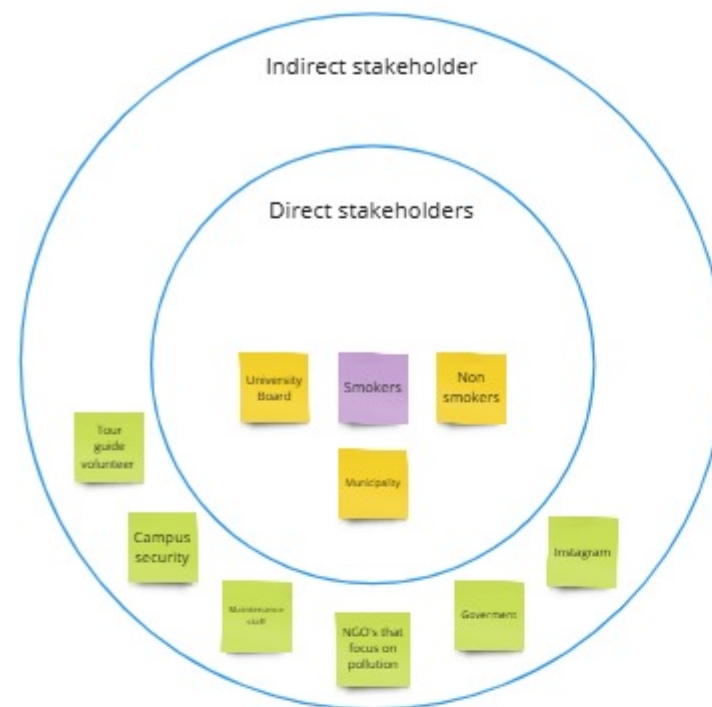
Simplified multibody analysis

- 1) list stakeholders:
- 2) list what each stakeholder can gain and lose
- 3) evaluate gains and losses in terms of values: **Stakeholder values map**
- 4) identify potential moral value conflicts
- 5) describe key principles/values that guiding to the analysis

Moral Value analysis

Accounting, equality, integrity, honesty, responsibility, cheerfulness, sharing, cooperation, freedom, identity, justice, membership, privacy, trust, authority, affiliation with one's community, emotional attachment, accumulating information, control, communication, accessibility, creating, reforming, enjoying, environmental sustainability, freedom of speech, thought, press, movement for community liberation.

1. Select relevant values.
2. Formulate the unitary results.
3. Consider having, doing, and being values.

[illegible]

Questionnaire: Bin the Butt campaign

1. This research project will be managed by Sanchayita Kainth, Helena Mossel, Tess Grobben & Chenwei Liang. The purpose of this questionnaire is to gain insights on how student smoke on campus and litter the cigarette butt. This data will be used to design a concept to legally allow students to smoke while keeping the environment clean. The contact data of TU/e are: Technische Universiteit Eindhoven, De Groene Loper 3, 5612 AE Eindhoven

You will be taking part in a research project in which we will gather information via a digital questionnaire. For your participation in this research project you will not be compensated.

All data will be anonymized.

Participation in this research is entirely voluntary. You do not have to answer questions you do not wish to answer. You may end your participation in the research project at any moment, or withdraw your consent to using your data for the research, without specifying any reason. Ending your participation will have no disadvantageous consequences for you. If you decide to end your participation during the research, the data which you already provided up to the moment of withdrawal of your consent will be used in the research.

Do you wish to end the research, or do you have any questions and/or complaints? Then please contact the researchers
s.kainth@student.tue.nl, h.t.mossel@student.tue.nl, t.h.e.grobben@student.tue.nl, c.liang1@student.tue.nl

If you have specific questions about the handling of personal data you can direct these to the data protection officer of TU/e by sending a mail to functionarisgegevensbescherming@tue.nl. Furthermore, you have the right to file complaints with the Dutch data protection authority: the Autoriteit Persoonsgegevens.

Finally, you have the right to request access, rectification, erasure or adaptation of your data. Submit your request via privacy@tue.nl.

By signing this consent form I acknowledge the following:

1. I am sufficiently informed about the research study. I have read the information and have had the opportunity to ask questions. These questions have been answered satisfactorily.
2. I take part in this research project voluntarily. There is no explicit or implicit pressure for me to take part in this research project. I am clear that I can end participation in this research project at any moment, without giving any reason. I do not have to answer a question if I do not wish to do so.

- ☐ I give consent
- ☐ I do not give consent

2. How often do you smoke cigarettes on campus?

- ☐ Daily
- ☐ Few times a week
- ☐ Few time a month
- ☐ Never

3. How often would you throw your cigarette butts on the floor?

- ☐ Never
- ☐ Sometimes
- ☐ Always

4. What is the reason for your choice for the above question?

5. Did you watch all the stories and posts uploaded on the Instagram page: Bin the Butt?

- ☐ Yes
- ☐ No

6. How would you describe the influence of those posts with your current behaviour?

7. What factors of the voting bins and smoking routes do you like or dislike? Please specify

8. How did you feel reading about the consequences of littering cigarette butts on the floor?

9. How did you feel about the posts suggesting you to go for a smoke walk on the suggested routes?

10. How interested are you in going for a smoking walk on the suggested routes?

Not at all

Somewhat Interested

Interested

Very Interest

☐☐☐☐

11. How excited are you to use the voting bins to dispose your cigarette butts? 1 being the least excited and 5 being the most excited

1	2	3	4	5
☐	☐	☐	☐	☐

12. Do you have comments and feedback about the educational social media campaign "Bin the Butt"

13. Do you have comments and feedback about the installation of voting bins and walking routes around campus where you can smoke freely?

This content is neither created nor endorsed by Microsoft. The data you submit will be sent to the form owner.



Responses Overview Active

Responses

3



Average Time

11:25



Duration

21 Days



1. This research project will be managed by Sanchayita Kainth, Helena Mossel, Tess Grobбен & Chenwei Liang. The purpose of this questionnaire is to gain insights on how student smoke on campus and litter the cigarette butt. This data will be used to design a concept to legally allow students to smoke while keeping the environment clean. The contact data of TU/e are: Technische Universiteit Eindhoven, De Groene Loper 3, 5612 AE Eindhoven

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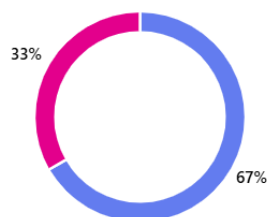
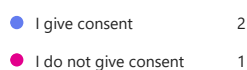
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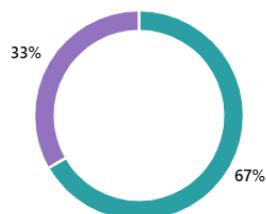
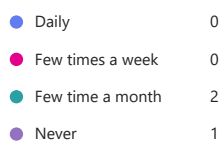
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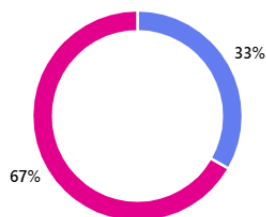


2. How often do you smoke cigarettes on campus?



3. How often would you throw your cigarette butts on the floor?

● Never	1
● Sometimes	2
● Always	0



4. What is the reason for your choice for the above question?

3
Responses

Latest Responses

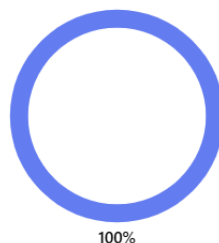
"Cigarette butts are bad for the environment and it takes little effort to throw them..."

"Not everywhere you can find a trashcan to put it in. Not even when you leave the..."

"Depends on the available bins nearby"

5. Did you watch all the stories and posts uploaded on the Instagram page: Bin the Butt?

● Yes	3
● No	0



6. How would you describe the influence of those posts with your current behaviour?

3
Responses

Latest Responses

"I wouldn't say they influence me much since I already throw my cigarette butts in..."

"It makes you think twice about the effects it might have"

"Made me more conscious, but at the same, I rarely throw them on the street. My..."

7. What factors of the voting bins and smoking routes do you like or dislike? Please specify

3
Responses

Latest Responses

"What I like about the smoking bins is that they stand out. They are easy to see wh..."

"Voting bins are a fun idea and makes it a game. The routes are a nice option, but..."

"I like the voting bins, it's creative, however the TUE campus is trying to become S..."

8. How did you feel reading about the consequences of littering cigarette butts on the floor?

3
Responses

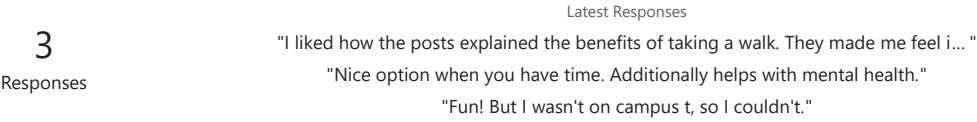
Latest Responses

"I think it's very useful information that would make people think twice before deci..."

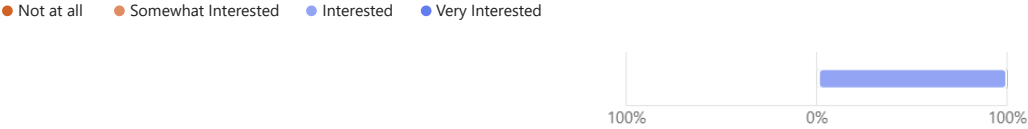
"This is hopefully a well known set of facts. When possible most people will put it in..."

"Nothing new..."

9. How did you feel about the posts suggesting you to go for a smoke walk on the suggested routes?



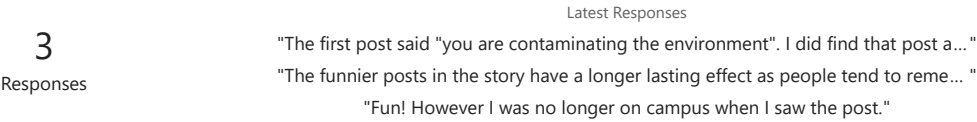
10. How interested are you in going for a smoking walk on the suggested routes?



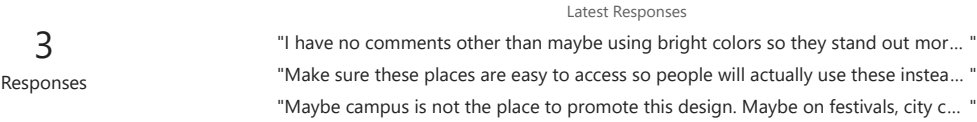
11. How excited are you to use the voting bins to dispose your cigarette butts? 1 being the least excited and 5 being the most excited



12. Do you have comments and feedback about the educational social media campaign "Bin the Butt"



13. Do you have comments and feedback about the installation of voting bins and walking routes around campus where you can smoke freely?



Ethical Review Form

(Version 2.1)

This Ethical Review Form should be completed for every research study that involves human participants or personally identifiable personal data and should be submitted to ethics@tue.nl. For more information about how this process works please click [here](#). Please check if you are using the correct form: Ethical Review Form (version 2.1). Please click [here](#) to obtain this latest version.

Part 1: General Study Information

1	Project title / Study name	DDM 110: Design for Behaviour Change: Bin the Butt
2	Name of the researcher / student	Sanchayita Kainth, Helena Mossel, Tess Grobben, Chenwei Liang
3	Email of the researcher / student	s.kainth@student.tue.nl , h.t.mossel@student.tue.nl , t.h.e.grobben@student.tue.nl , c.liang1@student.tue.nl
4	Supervisor(s) name(s) <i>Additional explanation: Please write down the name of your direct supervisor. You can mention several supervisors if appropriate, but at least one supervisor should be mentioned.</i>	Panos Markopoulos, Harm van Essen
5	Supervisor(s) email address(es) <i>Additional explanation: Please give the email address of the supervisor(s) mentioned in question 4.</i>	p.markopoulos@tue.nl , h.a.v.essen@tue.nl
6	Department / Group <i>Additional explanation: Please specify group if relevant e.g. JADS or HTI</i>	Industrial Design
7	What is the purpose of this application?	☐ Scientific study ☐ Bachelor education. Course:..... ☒ Master education. Course: DDM 110 Design for Behaviour Change ☐ Other (e.g. external, following external regulations):.....
8	Research location <i>Additional explanation: Where will the data collection take place? On campus, in a company, in public space, online, etc.</i>	☒ Eindhoven University of Technology campus ☐ Other, name organization(s):..... ☐ Public space ☐ Online
9	Start date data collection <i>Additional explanation: Please state when your data collection will start. Please note that you do not have to provide information about your complete (PhD) project, but only on this particular sub-study that you are submitting for approval in this form.</i>	September 2024
10	End date data collection	October 2024
11	Does your project receive external funding (e.g., NWO, relevant for special regulations from funders)?	☐ Yes. Name Funder: ☒ No

Ethical Review Form

12	Which internal and external parties are involved in the study? Think about sharing data or information between TU/e and other universities, commercial companies, hospitals, etc. <i>Additional explanation: Describe all internal and external parties that are involved in the study or project, including:</i> • researchers or research groups at the TU/e who participate in the study; • (Researchers at) other universities/institutions that provide data/services, help analyzing the data, etc.;	Internal parties • Researcher(s): Sanchayita Kainth, Helena Mossel, Tess Grobben, Chenwei Liang • Supervisor: Harm Van Essen, Panos Markopoulos
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Ethical Review Form

	(commercial) partners, companies, government bodies, municipalities, consultancy firms, hospitals or care institutions that provide data (e.g., contact details of participants, data for further analysis). Indicate which role each party plays: who defines the means and purposes in the study, who will supply the data (external parties?), who will process/handle the data, who will be able to access the data during and after research (only researchers at TU/e or also others)?	External parties - Other universities/institutions: - Others:N/A...
13	Have any special agreements already been made with an external party, such as a Non-Disclosure Agreement (NDA) or a data sharing agreement?	☐ Yes, namely: ☒ No
14	Has your proposal already been approved by an external Ethical Review Board or Medical Ethical Review Board? <i>Additional explanation:</i> For example, when you are collaborating with another university and the project has been approved by their Ethical Review Board, or when you received a WMO-waiver from a Medical Ethical Review Board.	☐ Yes ☒ No
15	If yes: Please provide the name, date of approval and contact details of the ERB. Please also include the registered number for your project approval. Additionally, please send in the Ethical Review Form upon which ethical approval was granted together with this form.	
16	If you process personal data that are likely to result in high privacy risks for participants, you need to perform a Data Protection Impact Assessment (DPIA). Have you done this for this or a very similar project? Please read the information below: a DPIA is not the same as a regular privacy impact assessment. More detailed questions on privacy will follow in the section below. <i>Additional explanation:</i> A Data Protection Impact Assessment (DPIA) is a formal document that must be drafted under the guidelines of the General Data Protection Regulation (GDPR). Think of research with vulnerable people, high-risk medical research, The Dutch DPA (Autoriteit Persoonsgegevens) and our website provides more information about a DPIA.	☒ Not applicable (no high privacy risks) ☐ Yes (the form is attached to the application) ☐ No
Part 2: Medical study		
1	Does the study have a medical scientific research question or claim? <i>Additional explanation:</i> Medical/scientific research is research which is carried out with the aim of finding answers to a question in the field of illness and health (etiology, pathogenesis, signs/symptoms, diagnosis, prevention, outcome or treatment of illness), by systematically collecting and analyzing data. The research is carried out with the intention of contributing to medical knowledge which can also be applied to populations outside of the direct research population. If your research contains questions about health and health related parameters (such as well-being, vitality, feelings of anxiety or stress) but your research question is not primarily medical, then you can answer 'no' to this question.	☐ Yes* ☒ No *If yes or in doubt, please contact Susan Hommerson via s.m.hommerson@tue.nl

Ethical Review Form

Part 3: Use of (medical) devices in the study

1	Does your research include a device? <i>Additional explanation: A device is a complete piece of physical hardware that is used to compute or support computer functions within a larger system. Devices can be divided into input-, output-, storage-, internet of things-, or mobile device.</i>	☐ Yes, not self-made ☐ Yes, self-made ☒ No
2	Please describe your device or link to an online description of the device	N/A
3a	Will you use a device that is 'CE' certified for unintended use (meaning you will use existing CE certified devices for other things than they were originally intended for) or use a device that is not 'CE' certified? <i>Additional explanation: You can find more information about CE certification here</i>	☐ Yes ☒ No ☐ ☐
3b	If no: Please explain to what extent the device was assembled according to relevant standards and provide a risk assessment <i>Additional explanation: You can find more information about a risk assessment here</i>	Microsoft Forms used to collect participant responses from surveys or semi structured interviews.
3c	If yes: Do you use a device or software that has a medical purpose such as diagnosis, prevention, monitoring, prediction, prognosis, treatment or alleviation of disease or injury?	☐ Yes, my device or software currently has a medical purpose ☐ Yes, my device or software could have a medical purpose in the near future ☒ No ☐ I'm not sure

Part 4: Information about the study

1	What are your main research questions? <i>Additional explanation: You need to provide at least one clear research question.</i>	How to change the behaviour of throwing cigarette butts on the floor causing littering, especially in a non-smoking area like TU/e campus
2a	Please check the box that indicates the relevant study population <i>Additional explanation: Please select which persons are eligible for your study.</i>	☒ Students ☐ General healthy population ☐ General population with specific feature, e.g., pregnancy, specifically ☐ Patients, specifically ☐ Other, specifically
2b	Age category of participants	☐ Younger than 12 years of age ☐ Older than 11 and younger than 16 years of age ☒ 16 years or older
3	Description of the research method (select all that applies)	☒ (Semi-structured) interviews ☒ Surveys

Ethical Review Form

	<u>Additional explanation:</u> Please specify your research method. Note that you need to provide information about the research method in an additional file that you attach to the ERB form. E.g., for interviews you provide the interview questions, for surveys you provide the survey questions, etc.	☐ Group workshops/roundtable discussions ☐ Diary studies ☒ Behavioral observations ☐ Building sensor data ☐ Wearable device (e.g. Fitbit watch, on-skin sensors) ☐ User testing ☒ Pilot study ☐ GPS tracking/location data ☐ Living Lab ☐ Other, namely
4	Description of the measurements and/or stimuli/treatments <u>Additional explanation:</u> Think about your outcome measures and the variables you will be collecting and describe them in a way such that another person understands what the participant will experience. For example: Participants will perform task A and see pictures from database B, and we measure validated Scale 1.	Participants will follow a social media Instagram page called "Bin the Butt", educational information about the consequences of littering cigarette butts will be shared to the participants through this Instagram page. Along with this our design concept and idea of voting bins/ashtrays installed outside the campus will also be shared. (Only photos and graphics of the bins will be shared, no physical voting bin/ashtray will be installed)
5	Describe and justify the number of participants you need for this study. Also justify the number of observations you need, taking into account the risks and benefits. <u>Additional explanation:</u> Think about if you need 3 or 30 participants for example, and why? Do they need to provide their input once, or several times, and why? If relevant, specify the duration of the study per participant and the compensation that is needed for the study.	Approximately, 4-5 participants to follow the Instagram page followed by informal interview or surveys
6	Explain why your research is societally important. What benefits and harm to society may result from the study? <u>Additional explanation:</u> What benefit will the results of your study have to society in general?	The goal is to change the behaviour of littering cigarette butts on the floor which have direct consequences to the environment and birds/animals.
7	Describe the way participants will be recruited <u>Additional explanation:</u> How will you recruit participants for your study? For example, by using flyers, personal network, panels, etc.	☒ Survey link posted online, e.g., social media platforms ☐ On campus flyers ☒ Personal network ☐ Via a company, namely ☐ Via a hospital, namely ☐ Via an organization ☐ By a Consortium Partner, namely ☐ Other, namely
8	Provide a brief statement of the risks you expect for the participants or others involved in the study and explain. Also take into consideration any personal data you may gather and associated privacy issues. <u>Additional explanation:</u> Risks for the participants can be anything from risk of data breach to risk of safety or well-being (think about stress, extreme emotions, visual or auditory discomfort). Describe these possible risks and describe the way these risks are mitigated.	No expected risk. No personal data will be collected. Only questions regarding the impact of educational posts and their attitude towards a designated legal smoking area outside campus.

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Part 5: Self-assessment checklist

Note: answers in the blue boxes indicate that your research is eligible for fast-track approval

		Yes	No
1a	Does the study involve human material? (e.g., surgery waste material derived from non-commercial organizations such as hospitals)		X
1b	Will blood or other (bio)samples be obtained from participants? (e.g., hair, sweat, urine or other bodily fluids or secretions, also external imaging of the body)		X
2	Will the participants give their consent – on a voluntary basis – either digitally or on paper? Or have they given consent in the past for the purpose of education or for re-use in line with the current research question?	X	
3	Are the participants, outside the context of the research, in a dependent or subordinate position to the investigator? Additional explanation: Think about doing research on your own students or on your own employees. When there is a dependency or power imbalance between you and the research participants, you need to answer 'yes' to this question.		X
4	Does the study involve participants who are particularly vulnerable or unable to give informed consent? (e.g., children (<16 years of age), people with learning difficulties, patients, people receiving counselling, people living in care or nursing homes, people recruited through self-help groups)		X
5	Will participating in the research be burdensome? (e.g., requiring participants to wear a device 24/7 for several weeks, to fill in questionnaires for hours, to travel long distances to a research location, to be interviewed multiple times)?		X
6	May the research procedure cause harm or discomfort to the participant in any way? (e.g., causing pain or more than mild discomfort, stress, anxiety or by administering drinks, foods, drugs, or showing explicit visual material)		X
7	Will financial inducement (other than reasonable expenses and compensation for time) be offered to participants? Additional explanation: For an explanation of what is considered a reasonable compensation, see the topic participant fees from the HTI group		X
8a	Will it be necessary for participants to take part in the study without their knowledge and consent at the time? (e.g., covert observation of people)		X
8b	If yes: Will you be observing people without their knowledge in public space? (e.g. on the street, at a bus-stop)		X
9	Will the study involve actively deceiving the participants? (e.g., will participants be deliberately falsely informed, will information be withheld from them, or will they be misled in such a way that they are likely to object or show unease when debriefed about the study)		X
10	Will participants be asked to discuss or report sexual experiences, religion, alcohol or drug use, suicidal thoughts, or other topics that are highly personal or intimate? Additional explanation: Think about your research population. For some participants, particular topics can be considered sensitive or intimate, whereas the same topics will not be perceived as such by other participants.		X
11	Elaborate on all boxes answered outside of the blue boxes in part 5. Describe how you safeguard any potential risk for the research participant.		

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Part 6: Self-assessment on privacy

The following questions (1-11) concern privacy issues, as laid down in the General Data Protection Regulation (GDPR). The Data Stewards and – if necessary – privacy team of TU/e will assess these questions. In some cases, more information is required to assess the privacy risks. If this is the case, you will be notified that the Data Stewards team will contact you.

The GDPR defines ‘personal data’ as any information relating to an identified or identifiable natural person (‘data subject’). Personal data also includes data that indirectly reveals something about a natural person. Personal data can lead to the physical, physiological, genetic, mental, economic, cultural or social identity of a natural person. There are two main categories of personal data: regular personal data and special category personal data.

If you are not sure whether some of these questions below should be answered with a Yes or No, please contact a Data Steward first through rdmsupport@tue.nl.

Note: answers in the blue boxes indicate that your research is eligible for fast-track approval

		Yes	No
1	Will the study involve discussion/collection/processing of regular personal data, or will you collect and (temporarily) store video or voice recordings for the purpose of conducting interviews? <i>Additional explanation:</i> For example, name, address, phone number, email address, IP address, gender, age, video or interview recordings? If you are not sure whether your data contains personal data, please contact the Data Stewards Team (rdmsupport@tue.nl).		X
1A	If yes: Please describe which regular personal data you will collect in this study?		
2	Will the study involve discussion/collection/processing of special category personal data or other sensitive data ? <i>Additional explanation:</i> Examples of special category personal data are race, religion, health information, political views, genetic or biometric data for the unique identification of a person, sexual preference, etc. Health information concerns personal data of the physical or mental health of persons, including the provision of health care. Examples of other sensitive data is information such as communication data, financial records or credit scores, camera surveillance data, location/GPS data, internet-of-things data, employee monitoring, observing or influencing behaviour, criminal records, <u>data of vulnerable persons (children, people with disabilities, refugees)</u> , BSN number etc. Please be aware that the use of special category personal data in research requires extra security measurements in order to safeguard the privacy of data subjects and to comply with the GDPR. Processing of this special category data is prohibited, except for specific purposes and under certain circumstances. If you need to process special category data, please consult the data stewards at rdmsupport@tue.nl .		X
2A	If yes: Please describe which special-category personal data and/or sensitive data you will collect in this study?		
If you answered yes to either question 1 or 2, please answer the questions below. If you answered no to both questions, you can skip this part and continue onto part 7. Also, if an answer to any of the following questions is ‘yes’, please contact a Data Steward at rdmsupport@tue.nl			
		Yes	No
3	Will your project involve the processing of personal data on a large scale ? <i>Additional explanation:</i> In general, any processing that involves more than 10.000 data subjects should be considered “large scale”. However, if the data of approximately 1000 persons (or more) are involved, the data processing may still be considered large scale. In that case, besides the number of persons involved in the study, one should also assess (i) the amount of data collected from these persons taking into account the type/risk level of the personal data, (ii) the duration of the data processing, (iii) the geographic scope or extent of the processing. For example, if you would collect and process data across several European countries with 10+ socio-economic data items of 1200 individual persons for several years in a row, that is likely “large-scale processing”. Other examples of a large-scale processing activity are: - Monitoring driving behavior of road users on Dutch highways - Collecting data of Covid patients - A hospital that processes patient data as part of its usual operations		X

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	A transport company that processes travel information of people who travel by public transport in a certain city. For example, by tracking them through travel maps.		
4	Does this processing activity involve the use of new or innovative technologies? <i>Examples of a new technology: combining fingerprints and facial recognition for physical access control, the use of bodycams in public spaces, the use of new technical methods in conducting research such as AI. This question also refers to new technologies that have not been deployed by TU/e so far.</i>		X
5	Does your study involve systematic (c.q. automated) monitoring of persons? <i>Additional explanation: Consider data processing activities that have the purpose of observing, monitoring or controlling individuals, for example in circumstances where the individuals are not aware by whom their personal data is collected and how it is used. Examples of such activities are using camera systems to monitor driving behavior on highways, monitoring email inactivity or employee phone use, certain applications of machine learning and artificial intelligence.</i>		X
6	Does the study involve collaborations (with third parties) in which data are shared or exchanged in order to link or combine data? <i>Additional explanation: This may often apply in a collaboration between the university and a commercial party, contract research, etc. It is important to assess this for all data in the entire project, not just your own data. An important consideration in this situation is whether the person whose data is involved could have expected that data from these different databases or sources of information were to be combined. For example, it is less likely for data subjects to expect that databases from different parties will be combined and the results are used for different purposes than one could reasonably expect; this may apply for example in a collaboration between the university and a commercial party.</i>		X
7	Will the study include data processing activities that prevent data subjects from exercising their rights or using a service or contract? <i>Additional explanation: Examples include processing operations carried out in public places that people cannot avoid (train station, airport, shopping mall, public university premises, etc.) or processing operations whose purpose is to allow or not allow data subjects to use a service or enter into a contract (examples: by refusing to pay a benefit, not being able to apply for a loan, etc.).</i>		X
8	Will the study process personal data to score, rank or profile persons? <i>Additional explanation: Examples: monitoring (highway) roads to give road users a "score" based on their detected driving behavior, a bank assessing its customers based on their creditworthiness, or an organization building behavioral and marketing profiles based on use of their website or navigating their website.</i>		X
9	Does your data processing include activities that involves composing "blacklists" – and, in particular, in relation to sensitive or special category data, such as communication data, financial records or credit scores, genetic data, biometric data, health data, camera surveillance data, location/GPS data, internet-of-things data, employee monitoring, observing or influencing behaviour, etc. <i>Additional explanation: This situation will not be a common occurrence in research, but you may indirectly be involved in this. In general, this typically concerns processing operations involving personal data relating to criminal convictions and offences, data relating to unlawful acts, data concerning unlawful or annoying behaviour or data concerning bad payment behaviour by companies or individuals are processed and shared with third parties (blacklists or warning lists, as used, for example, by insurers, hospitality companies shopping companies, telecom providers as well as blacklists relating to unlawful behavior of employees, for example in the healthcare sector or by employment agencies, etc.).</i>		X
10	Will personal data be transferred or shared outside the EU/EEA? EU data protection rules apply to the European Economic Area (EEA), which includes all EU countries and non-EU countries Iceland, Liechtenstein and Norway. <i>Additional explanation: The GDPR has drafted additional requirements for transfers data outside of the EU/EEA. Typically, additional safeguards must be implemented to protect the personal data of residents in the European Union. For example, if you collaborate with an American, Indian or Chinese university or other third party outside the EU/EEA, you must first check whether this is allowed and under which conditions this is allowed. Another typical example is storage of data on American providers of cloud (storage) services. Please contact the data stewards first to discuss this.</i>		X
11	Will any raw or anonymized personal data or any other sensitive data or research results from the project possibly be transferred to a high-risk country*? *High risk countries: China, Russia, Iran, Turkey, and North Korea. <i>If personal data or other potentially sensitive data is exchanged with one of these countries, or if part of the data processing takes place in one of these countries: an advice from the Data Protection Officer, the kennisveiligheidsteam (Knowledge Security team), and the CISO (Chief Information Security Officer) is ALWAYS required.</i>		X

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Part 7a: Processing of research data

1	Is consent your legal basis for processing the personal data in your study? <i>Additional explanation: What is a legal basis? One of main principles in the GDPR is to ensure that personal data is processed lawfully, fairly, and transparently. To comply with this principle, the processing of personal data also requires that you have a valid legal basis for the personal data processing activity.</i> <i>In research projects, the legal basis is often but not always consent. However, it is possible that it is not clear or not possible to establish whether to use consent as a legal basis.</i> <i>Some examples where consent may not be applicable as legal basis are covert research, data collection in public spaces, secondary data analysis of existing data, data that are transferred to you by a third party, consent is not possible or would require disproportionate effort, etc. In that case, please indicate which legal basis you think that applies or (preferably) contact a data steward first.</i>	☒ Yes and it will be obtained via CONSENT FORM QUESTION (Before the beginning of the survey) An informed consent template* is attached to this application. ☐ No, I will use another legal basis to process the data. Namely, * You can download a suitable template here.
2	Where will the data come from?	☐ Data obtained from another party (secondary data use) ☒ New data collected only by my research team ☐ New data collected together with collaborators
3	Which of the following tools will you use to process personal data?	Surveys ☐ Qualtrics ☐ Limesurvey ☒ MS Forms ☐ Other, namely Interview/workshop recordings ☐ Voice/video recorder ☐ Phone in a flight mode ☐ MS Teams ☐ Other, namely Transcription ☐ Manual transcription ☒ Microsoft Office software (e.g. Word, Teams) ☐ Other, namely Statistical analysis ☐ SPSS ☐ R ☐ Other, namely Other tools, specifically.....
4	Where will the data and in particular the personal data be stored during and after completion of the study? If you have already uploaded your Data Management Plan, you can refer to your Data Management Plan.	☐ SURF drive ☒ Onedrive ☐ Research Drive ☐ Network Drive

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	<i>Additional explanation: University supported-storage facilities are SURFdrive, SURF Research Drive, Ceph, departmental drives (this includes BE Project Drive), and the TU/e instance of Microsoft OneDrive. For most personal data, the use of SURF Research Drive, departmental drives (including BE Project Drive) and SURFdrive is required.</i>	☐ Research Manager ☐ Other, namely
Part 7b: Safety and security measures		
1	Will you pseudonymize/anonymize the data? <i>Additional explanation:</i> <i>Anonymization: remove all direct identifiers (name, address, telephone number etc.) but also indirect identifiers (age, place of birth, occupation, salary) that, linked with other information, can lead to a person's identification. Anonymization to the point that a data subject is no longer identifiable means that the anonymized data is not considered to be personal data anymore.</i> <i>Pseudonymization: replacing the unique identifier of a data subject with an artificial pseudonym. This means that identification is still possible with the identification key. The identification key needs to be stored securely and separately from the pseudonymized data. If the data subject can be identified by combining data with additional information, the data is also called pseudonymous.</i>	☒ Yes ☐ No If yes, describe how: No names or other personal information will be collected. Each participant will be anonymized by giving them a participant number.
2	Is access to (personal) data restricted? (Select all that apply)	☐ No ☐ Yes, via access control ☐ Yes, via password protection ☒ Yes, access only given to TU/e research team ☐ Yes, access only given to research team, including non-TU/e collaborators ☐ Other, specify.....
3	Who will have access to the data during and after completion of the project? (Select all that apply)	☒ Main researcher ☒ TU/e supervisor(s) ☐ External supervisors ☐ TU/e research team ☐ Other, specify.....
4	Will you store data for future research?	☒ No ☐ Yes, in a public data repository ☐ Yes, in a public data repository under restricted access ☐ Yes, in a TU/e-recommended storage (SURF Research Drive, Network Drive)
5	Will you share data outside the TU/e?	☒ No ☐ Yes, in a fully anonymized form ☐ Yes, raw or pseudonymized data* *If you selected this box, make sure that a suitable data agreement is put in place. You can contact the Data Stewards for support in preparing such an agreement
6	How long will data be stored after the end of the project?	The data will be stored in the one drive folder of the researchers and will be deleted by the researchers after the end of the project.

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Part 8: Closures and Signatures

1	Enclosures (tick if applicable and attach to this form):	☐ Informed consent form ☐ Informed consent form for other agencies when the research is conducted at a location (such as a school) ☐ Text used for ads (to find participants) ☐ Text used for debriefings ☐ Approval other research ethics committee ☒ The survey the participants need to complete, or a description of other measurements ☐ Data Protection Impact Assessment checked by the privacy officer ☐ Data Management Plan checked by a data steward
2	Signature(s)	Signature(s) of applicant(s) Skainth. Chenwoi   Date: 22/10/2024 Signature research supervisor Date: